

Geometry-Decorrelation of GNSS Carrier-Phase Gestures — Theory

Project: GNSS-based gesture recognition (FineSat-style), u-blox EVK-F9P **Question this derivation answers:** *How long is the time window over which the same hand gesture, performed in front of a fixed antenna, produces a reproducible carrier-phase signal — given that the satellite geometry drifts continuously?*

The headline theoretical result is that the window is governed by a single dimensionless curvature κ that depends on the **shape of the gesture**, and that **single-axis gestures are geometrically robust ($\kappa \rightarrow 0$) while multi-axis gestures decorrelate quadratically with geometry drift**.

1. Setup and signal model

A hand moves near a fixed GNSS antenna, acting as a moving multipath reflector. The antenna does **not** translate; the gesture is the motion of the reflector. Let

- $\mathbf{d}(t) \in \mathbb{R}^3$ — the hand displacement trajectory (the gesture), defined over a short window $t \in [0, T_g]$ (a few seconds),
- $\hat{\mathbf{e}}_i$ — the line-of-sight (LOS) unit vector from antenna to satellite i ,
- λ — the carrier wavelength.

To leading order, the multipath-induced carrier-phase perturbation on satellite i is proportional to the projection of the hand displacement onto the LOS direction:

$$\phi_i(t) \approx \frac{2\pi}{\lambda} \hat{\mathbf{e}}_i \cdot \mathbf{d}(t) + \underbrace{c(t)}_{\text{rx clock}} + \underbrace{\rho_i(t)}_{\text{geometric range trend}} + \text{noise}.$$

Single-differencing against a reference satellite removes the receiver clock $c(t)$ (common to all satellites), and a low-order **detrend** over the window removes the smooth geometric range trend $\rho_i(t)$. What remains is the gesture signal:

$$\boxed{s_i(t) = \mathbf{g}_i \cdot \mathbf{d}(t)}, \quad \mathbf{g}_i = \frac{2\pi}{\lambda} (\hat{\mathbf{e}}_i - \hat{\mathbf{e}}_{\text{ref}}).$$

$\mathbf{g}_i$ is the **differential LOS sensitivity vector**. The whole geometry of the problem enters through these vectors.

Model caveat (established empirically). The LOS-projection model is leading-order. Real specular reflection off the hand carries reflection-geometry factors (the sensitivity is along the bisector of incident/reflected rays, not pure LOS), and the antenna phase-center variation is not captured. In our data, a pure 3-D translation model explained ~69 % of the push variance but only ~39 % of the star variance, so the model below is a first-order description, not exact.

2. Correlation under geometry drift

Perform the *same* gesture $\mathbf{d}(t)$ at two times A and B separated by Δt . Between them the LOS rotates, so $\mathbf{g}_i^A \rightarrow \mathbf{g}_i^B$. The reproducibility of satellite i 's signal is the normalized zero-lag correlation over the window:

$$r = \frac{\langle s_A, s_B \rangle}{\|s_A\| \|s_B\|} = \frac{\int_0^{T_g} (\mathbf{g}_A \cdot \mathbf{d})(\mathbf{g}_B \cdot \mathbf{d}) dt}{\sqrt{\int (\mathbf{g}_A \cdot \mathbf{d})^2 dt} \sqrt{\int (\mathbf{g}_B \cdot \mathbf{d})^2 dt}}.$$

Each inner product is a quadratic form in the **gesture structure matrix** (the 3×3 second-moment / scatter matrix of the trajectory):

$$\boxed{M = \int_0^{T_g} \mathbf{d}(t) \mathbf{d}(t)^\top dt} \quad \Rightarrow \quad \int (\mathbf{g}_A \cdot \mathbf{d})(\mathbf{g}_B \cdot \mathbf{d}) dt = \mathbf{g}_A^\top M \mathbf{g}_B.$$

So the correlation is the **generalized cosine** between the two geometry vectors in the metric M :

$$r = \frac{\mathbf{g}_A^\top M \mathbf{g}_B}{\sqrt{\mathbf{g}_A^\top M \mathbf{g}_A} \sqrt{\mathbf{g}_B^\top M \mathbf{g}_B}}.$$

The entire dependence on the **gesture shape** is in M ; the entire dependence on **geometry drift** is in $\mathbf{g}_A \rightarrow \mathbf{g}_B$.

3. Small-drift expansion $\rightarrow$ the curvature κ

Write the drift as $\mathbf{g}_B = \mathbf{g}_A + \Delta \mathbf{g}$ with $\Delta \mathbf{g}$ small. Define the three scalars

$$a = \mathbf{g}_A^\top M \mathbf{g}_A, \quad b = \mathbf{g}_A^\top M \Delta \mathbf{g}, \quad c = \Delta \mathbf{g}^\top M \Delta \mathbf{g}.$$

Then $r = \frac{a+b}{\sqrt{a} \sqrt{a+2b+c}} = \frac{1+\beta}{\sqrt{1+2\beta+\gamma}}$ with $\beta = b/a$ (order $\Delta\theta$) and $\gamma = c/a$ (order $\Delta\theta^2$). Expanding to second order:

$$\sqrt{1+2\beta+\gamma} \approx 1 + \beta + \frac{\gamma}{2} - \frac{\beta^2}{2}, \quad r \approx (1+\beta) \left(1 - \beta - \frac{\gamma}{2} + \frac{3\beta^2}{2} \right) = 1 - \frac{1}{2}(\gamma - \beta^2).$$

Therefore

$$1 - r \approx \frac{1}{2} \frac{c a - b^2}{a^2}.$$

The LOS rotates by a small angle $\Delta\theta$, so $\Delta \mathbf{g} = \Delta\theta \mathbf{w}$ where $\mathbf{w} = d\mathbf{g}/d\theta$ is fixed by the geometry. Substituting $b = \Delta\theta (\mathbf{g}^\top M \mathbf{w})$ and $c = \Delta\theta^2 (\mathbf{w}^\top M \mathbf{w})$:

$$\boxed{r \approx 1 - \frac{1}{2} \kappa \Delta\theta^2}, \quad \boxed{\kappa = \frac{(\mathbf{w}^\top M \mathbf{w})(\mathbf{g}^\top M \mathbf{g}) - (\mathbf{g}^\top M \mathbf{w})^2}{(\mathbf{g}^\top M \mathbf{g})^2}}.$$

κ is dimensionless and is the **decorrelation curvature**.

3.1 Properties of κ

- **Non-negativity.** By the Cauchy–Schwarz inequality in the M -inner-product, $(\mathbf{g}^\top M \mathbf{w})^2 \leq (\mathbf{g}^\top M \mathbf{g})(\mathbf{w}^\top M \mathbf{w})$, so $\kappa \geq 0$ always. Correlation can only decrease with drift.
 - **$\kappa = 0$ condition.** $\kappa = 0$ iff $M \mathbf{w} \parallel M \mathbf{g}$, i.e. the drift direction is indistinguishable from the original direction *as seen through the gesture metric M* .
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4. The headline result: gesture rank sets the window

4.1 Single-axis gesture $\rightarrow \kappa = 0$ (geometry-robust)

If the gesture is one-dimensional, $\mathbf{d}(t) = f(t) \hat{\mathbf{n}}$ for a fixed direction $\hat{\mathbf{n}}$, then

$$M = \left(\int f^2 dt \right) \hat{\mathbf{n}} \hat{\mathbf{n}}^\top \quad (\text{rank } 1).$$

For any vector $\mathbf{x}$, $M\mathbf{x} = (\int f^2)(\hat{\mathbf{n}} \cdot \mathbf{x}) \hat{\mathbf{n}}$ — always parallel to $\hat{\mathbf{n}}$. Hence $M\mathbf{g} \parallel M\mathbf{w}$, and substituting into κ :

$$\kappa_{1D} = \frac{(\int f^2)^2 [(\hat{\mathbf{n}} \cdot \mathbf{w})^2 (\hat{\mathbf{n}} \cdot \mathbf{g})^2 - (\hat{\mathbf{n}} \cdot \mathbf{g})^2 (\hat{\mathbf{n}} \cdot \mathbf{w})^2]}{(\int f^2)^2 (\hat{\mathbf{n}} \cdot \mathbf{g})^4} = 0.$$

A single-axis gesture has no second-order geometric decorrelation. Intuition: a 1-D gesture senses only the scalar projection $\hat{\mathbf{n}} \cdot \mathbf{g}$. As geometry drifts, that projection changes *amplitude* but the signal *shape* $f(t)$ is unchanged — and correlation is amplitude-invariant. The window is then limited by reproducibility and higher-order terms, not geometry.

4.2 Multi-axis gesture $\rightarrow \kappa > 0$ (tighter window)

If $\mathbf{d}(t)$ explores two or three dimensions (a star, a circle), M is full-rank with spread eigenvalues, $M\mathbf{g}$ and $M\mathbf{w}$ are generally not parallel, and $\kappa > 0$. The signal *shape* — not just amplitude — changes with geometry, so correlation falls quadratically. The larger the eigenvalue spread of M (the more genuinely multi-dimensional the gesture), the larger κ and the tighter the window.

4.3 Choice of probe gestures

We use two gestures at the κ -extremes so everything else interpolates between them:

gesture	character	structure matrix eigenvalues (normalized)	κ	window
push	near single-axis	$\approx [1, 0.009, 0.008]$ ($\approx \text{rank-1}$)	≈ 0	wide
star	genuinely multi-axis	$\approx [1, 0.34, 0.28]$	> 0	tight (binding constraint)

(Eigenvalues are illustrative estimates; the operative values are the **empirical** M measured from the recovered data.)

5. The usable window $\delta t_{\max}$

Define the window by a minimum acceptable correlation $r_{\min}$:

$$r_{\min} = 1 - \frac{1}{2} \kappa \Delta \theta_{\max}^2 \implies \Delta \theta_{\max} = \sqrt{\frac{2(1 - r_{\min})}{\kappa}}.$$

The differential LOS rotates at an apparent rate ω_{LOS} , so $\Delta \theta = \omega_{\text{LOS}} \delta t$ and

$$\boxed{\delta t_{\max} = \frac{1}{\omega_{\text{LOS}}} \sqrt{\frac{2(1 - r_{\min})}{\kappa}}}.$$

Scaling consequences:

- $\delta t_{\max} \propto \kappa^{-1/2}$. Push ($\kappa \rightarrow 0$) $\rightarrow$ window limited by reproducibility, not geometry. Star (finite κ) $\rightarrow$ finite window.

- The push/star window ratio is $\sqrt{\kappa_{\text{star}}/\kappa_{\text{push}}}$, set entirely by the empirical M .
- $\omega_{\text{LOS}} \approx 0.45^\circ/\text{min}$ for GPS/BeiDou-MEO at this site (tightly clustered 0.42–0.52 across elevations, because the rate is dominated by orbital angular velocity, not elevation).

6. The reproducibility floor α (why mechanical reproduction matters)

At **zero** geometry drift, two free-hand repetitions are still not identical:

$$r(\Delta\theta = 0) = \alpha < 1,$$

the **reproducibility floor**, set by how precisely the human (or actuator) repeats the gesture, bounded above by the thermal/carrier-phase **noise ceiling** $r_{\text{max}} \approx 0.85$. A useful combined model is

$$r(\delta t) \approx \alpha \left(1 - \frac{1}{2} \kappa \omega_{\text{LOS}}^2 \delta t^2 \right).$$

The geometric window is only **measurable** when α is high enough that the quadratic geometry term — not free-hand sloppiness — dominates the falloff. Our measurements gave provisional free-hand $\alpha \approx 0.6$ (single-differenced, recovered data), well below the ~ 0.85 ceiling — which is why characterizing the *geometric* window requires **mechanically reproduced** gestures to isolate κ from human irreproducibility. The eventual free-hand product does not need signal correlation at all; it needs a classifier trained across geometry.

7. Geometry repeat (the direct measurement)

Rather than wait for natural drift, the geometry can be revisited. GPS/BeiDou-MEO geometry repeats every **sidereal** day:

$$T_{\text{sid}} = 86164.0905 \text{ s} \quad (\approx \text{solar day} - 3 \text{ m } 56 \text{ s}).$$

- **GPS only:** geometry recurs at $T + k T_{\text{sid}}$ (1 sidereal day; $\sim 3 \text{ m } 56 \text{ s}$ earlier each solar day).
- **GPS + BeiDou jointly:** GPS repeats in 1 sidereal day (2 orbits), BeiDou-MEO in 7 sidereal days (13 orbits); the **joint** constellation geometry repeats every **7 sidereal days** ($\approx T + 7 \text{ d} - 27 \text{ m } 31 \text{ s}$).

Direct window measurement (the repeat-offset sweep): collect a reference at T , then revisit at $T + T_{\text{sid}} + \varepsilon$ for a set of offsets ε . The induced drift is $\Delta\theta = \omega_{\text{LOS}} \varepsilon$, so sweeping ε traces $r(\Delta\theta)$ directly and the measured falloff is compared against the predicted $1 - \frac{1}{2} \kappa \Delta\theta^2$.

8. Summary of the theoretical result

1. The reproducible gesture signal is $s_i(t) = \mathbf{g}_i \cdot \mathbf{d}(t)$; correlation under drift is the M -metric cosine of the geometry vectors, $M = \int \mathbf{d} \mathbf{d}^\top dt$.
2. Decorrelation is quadratic: $r \approx 1 - \frac{1}{2} \kappa \Delta\theta^2$ with the closed-form curvature $\kappa = \frac{(\mathbf{w}^\top M \mathbf{w})(\mathbf{g}^\top M \mathbf{g}) - (\mathbf{g}^\top M \mathbf{w})^2}{(\mathbf{g}^\top M \mathbf{g})^2} \geq 0$.
3. **Single-axis gestures are geometry-robust** ($\kappa = 0$, rank-1 M); **multi-axis gestures decorrelate** ($\kappa > 0$). Gesture *shape*, via the rank/eigenvalue-spread of M , sets everything.
4. Window: $\delta t_{\text{max}} = \omega_{\text{LOS}}^{-1} \sqrt{2(1 - r_{\text{min}})}/\kappa$, with $\omega_{\text{LOS}} \approx 0.45^\circ/\text{min}$.
5. The measurement requires a reproducibility floor α near the noise ceiling; free-hand $\alpha \approx 0.6$ means **mechanical reproduction** is needed to isolate the geometric κ . Direct verification is via the sidereal **repeat-offset sweep**.